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The Short-Termism Trap: Catering to Informed Investors with Limited Horizons

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1 Big picture

- **Question.** Does the stock market exert short-term pressure on listed firms, do firms respond to that pressure, and can the response destroy shareholder value?
- **Main mechanism.** Informative stock prices improve managerial incentives. However, informed investors have limited horizons and prefer firms whose value is revealed quickly. Listed firms therefore choose shorter-maturity projects to attract informed trading.
- **Central externality.** Informed trading capital is scarce. When one firm shortens project maturity, it attracts informed investors away from other firms. Each firm ignores the informational loss imposed on others.
- **Short-termism trap.** In equilibrium, firms compete for informed investors by shortening project maturity. Because informed capital is fixed, this competition does not raise aggregate price informativeness; it just lowers project value and destroys shareholder surplus.
- **Listing decision.** Public listing is valuable because stock prices can discipline managers. But investor short-termism can become so strong that some firms choose to remain unlisted.
- **Equilibrium result.** There is a unique equilibrium. Listed firms choose symmetric interior maturities, and if investor short-termism is high enough, only a subset of firms list.
- **Benchmark comparison.** Equilibrium project maturity is shorter than the coordinated benchmark. Shareholder value is lower than under coordinated maturity choices.
- **Comparative statics.** More competition among listed firms, stronger investor short-termism, and stronger managerial agency problems can all worsen the short-termism trap.
- **Long-term investors.** Introducing long-term investors can help, but does not necessarily eliminate the trap. When clienteles form, some firms choose short projects and others choose longer projects.
- **Policy message.** Restrictions aimed directly at executive compensation or project choice may have ambiguous effects. The root problem is an informational externality in competition for informed trading.
- **Takeaway.** Stock market informativeness is socially useful, but individual firms can over-cater to short-horizon informed investors. Market monitoring can therefore create a race to the bottom in project maturity.

2 Model primitives and measurement

2.1 Firms and project maturity

- There are M firms, indexed by $n = 1, \dots, M$.
- A subset $\mathcal{N}$ of N firms is listed and traded in the stock market.
- Shareholders choose a project maturity $\tau_n \in [0, 1]$.
- The project matures early at date 1 with probability $1 - \tau_n$ and late at date 2 with probability τ_n .

The project payoff is

$$V^n = f(\tau_n) + R_n, \tag{1}$$

where

$$R_n = \begin{cases} \Delta V(1 + \alpha \mathbf{1}_\ell), & \text{if the project is successful,} \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Interpretation: The function $f(\tau_n)$ is the maturity-sensitive production payoff. It is increasing and concave: long projects are more productive, but the marginal benefit of maturity eventually falls.

2.2 Managerial effort and signals

- Each manager chooses high or low effort, $e \in \{H, L\}$.
- High effort has private cost K .
- Informed investors who investigate firm n receive a signal $s_n \in \{G, B\}$.
- A good signal is more likely under high effort:

$$\sigma_H = \Pr(s_n = G \mid e_n = H), \quad \sigma_L = \Pr(s_n = G \mid e_n = L), \quad \Delta\sigma = \sigma_H - \sigma_L > 0. \quad (3)$$

Let ν_G and ν_B denote the posterior probabilities of a high payoff conditional on good and bad signals. Then

$$\rho_H = \sigma_H \nu_G + (1 - \sigma_H) \nu_B, \quad \rho_L = \sigma_L \nu_G + (1 - \sigma_L) \nu_B. \quad (4)$$

Interpretation: The stock price can be useful in contracts because it partially reveals whether the manager exerted high effort.

2.3 Informed investors and short horizons

- There is a unit mass of risk-neutral informed investors.
- They can acquire private information about one listed firm.
- With probability γ , an informed investor must consume early and therefore has a short horizon.
- With probability $1 - \gamma$, the investor can hold until date 2.

Interpretation: The parameter γ measures investor short-termism. A higher γ means informed investors value projects whose information is revealed sooner.

2.4 Price informativeness

Let λ_n denote price informativeness for listed firm n .

- More informed investors following firm n increases λ_n .
- Shorter project maturity increases the benefit to short-horizon informed investors, attracting them to firm n .
- Because informed capital is scarce, attracting investors to one firm reduces informativeness for other firms.

In a symmetric equilibrium with N listed firms, price informativeness is

$$\lambda_n = \frac{1}{N\bar{z}}, \quad \forall n \in \mathcal{N}, \quad (5)$$

where $\bar{z}$ captures noise trading intensity relative to informed trading.

Interpretation: When all listed firms shorten maturity together, no individual firm gains informativeness because the total mass of informed investors is fixed.

2.5 Shareholder value

Shareholder value is project output net of the managerial wage bill:

$$S_n = V^n - w_n. \quad (6)$$

For listed firms that induce high effort, equilibrium shareholder value has the form

$$S_n^* = f(\tau_n^*) + \rho_H \Delta V - [\lambda_n \sigma_H w_G^* + (1 - \lambda_n)(1 - \delta \tau_n^*) \rho_H w_S^*]. \quad (7)$$

Interpretation: A shorter maturity reduces the production term $f(\tau)$ but may reduce the compensation cost by making prices more informative.

3 Models and theoretical results

3.1 Financial market equilibrium

Proposition 1. Given project maturities $\{\tau_n\}_{n \in \mathcal{N}}$, there exists a unique financial market equilibrium. Price informativeness λ_n is decreasing in firm n 's own maturity and increasing in other firms' maturities.

Interpretation: A firm can attract informed investors by shortening its project maturity. But this steals informed capital from other firms.

3.2 Optimal managerial contract

The contract pays the manager according to the information in the stock price. Let w_G be compensation after good price information and w_S be compensation after successful final output in states where price information is uninformative.

The incentive constraint is

$$\lambda_n \Delta \sigma u(w_G) + (1 - \lambda_n)(1 - \delta \tau_n) \Delta \rho u(w_S) = K. \quad (8)$$

The expected wage bill is

$$\mathcal{W}_n = \lambda_n \sigma_H w_G + (1 - \lambda_n)(1 - \delta \tau_n) \rho_H w_S. \quad (9)$$

Proposition 2. Given τ_n and λ_n , the optimal contract minimizes the expected wage bill subject to the incentive constraint. The wage bill is increasing in project maturity, holding price informativeness fixed.

Interpretation: Longer projects make the stock price less useful for incentivizing managers, so shareholders must pay a higher wage bill.

3.3 Optimal maturity choice

A listed firm's optimal maturity satisfies a first-order condition equating marginal production benefits and marginal compensation-cost effects:

$$\text{Marginal gain in production from higher } \tau_n = \text{Marginal increase in the wage bill from lower price informativeness.} \quad (10)$$

Proposition 3. Given other firms' maturity choices, each listed firm has an optimal project maturity.

Interpretation: A firm chooses maturity by balancing a real production benefit from longer projects against a contracting benefit from shorter projects.

3.4 Strategic complementarity

Proposition 4. A firm's optimal maturity choice is increasing in other firms' maturity choices:

$$\frac{\partial \tau_n^*}{\partial \tau_m} > 0, \quad m \neq n. \quad (11)$$

Interpretation: If one firm shortens maturity, it attracts informed investors and lowers other firms' price informativeness. Other firms respond by shortening maturity as well. This is the race-to-the-bottom force.

3.5 Listing decision and equilibrium

Unlisted firms do not use stock prices for incentive contracting. They choose the long project and obtain

$$S^U = f(1) + \rho_L \Delta V. \quad (12)$$

A listed firm lists if $S_n^* \geq S^U$.

Theorem 1. There exists a unique equilibrium. There is a critical value γ^* for investor short-termism such that all firms list if $\gamma \leq \gamma^*$, whereas some firms remain unlisted otherwise. Listed firms choose a symmetric interior maturity.

Interpretation: Listing can lose value when investor short-termism is high. Some firms then prefer not to be publicly traded.

3.6 Benchmark cases and inefficiency

The paper compares equilibrium to three benchmarks:

1. effort without stock prices;
2. coordinated maturity choice among listed firms;
3. equilibrium maturity choice under competition for informed trading.

Theorem 2. In the coordinated benchmark, firms choose a longer maturity than in equilibrium.

Theorem 3. Shareholder value is lower in equilibrium than in the coordinated benchmark:

$$S^{EP} < S^* < S^{CB}. \quad (13)$$

Interpretation: Firms are individually rational but collectively inefficient. They over-shorten projects because they do not internalize the negative information externality imposed on other firms.

3.7 Comparative statics and extensions

- **Competition.** More firms competing for the same informed capital can increase short-termism.
- **Investor short-termism.** A higher γ reduces equilibrium maturity and shareholder value.
- **Agency problem.** A stronger managerial agency problem can amplify the trap.
- **Long-term investors.** A mass of long-term investors can support longer maturity choices but may lead to clientele equilibria.
- **Salary caps.** A salary cap does not necessarily lengthen maturity; it can worsen the short-termism trap by weakening incentive provision.

Interpretation: The trap is robust because it is driven by an externality in the market for informed trading, not only by executive compensation design.

4 Identification and empirical implications

4.1 Observable predictions

Although the paper is theoretical, it implies several empirical predictions:

- Firms with more short-horizon investor bases should choose projects with faster payoff realization.
- Industries with more competition for investor attention should display more short-term project choices.
- Greater reliance on stock-price-based incentives should strengthen the pressure to shorten maturity.
- Long-term investor clienteles should be associated with longer project horizons and lower price informativeness.

Interpretation: The model suggests looking for a link between investor horizons, price informativeness, managerial compensation, and project maturity.

4.2 Relevant empirical settings

- Firms with high institutional turnover or hedge-fund ownership.
- Firms that choose between short-cycle and long-cycle investment projects.
- Periods in which index inclusion, liquidity, or analyst coverage changes informed trading attention.
- Governance reforms that change the role of stock prices in executive compensation.

Interpretation: The key test is not simply whether short-term investors exist, but whether firms alter real project maturity to attract informative trading.

4.3 Limitations

- The model is stylized and abstracts from many real investment frictions.
- The source of investor short horizons is taken as given in the baseline.
- The model focuses on stock-price informativeness as a contracting device, though stock prices can affect firms in other ways.
- The welfare analysis is mainly about shareholder value rather than broader stakeholder welfare.

Interpretation: The abstraction isolates a clean information externality: firms compete for informed trading even when doing so reduces collective value.

5 Tables and figures

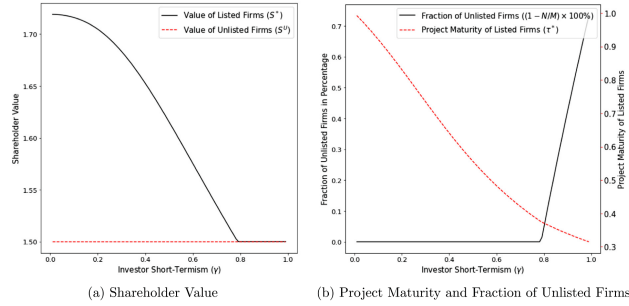
The paper is theoretical and contains two main figures. The figures below are directly cropped from the paper and grouped with concise readings.

5.1 Figure 1: Investor short-termism, listing decisions, and shareholder value

Read:

- Panel A plots the shareholder value of listed firms and unlisted firms as investor short-termism γ increases.
- Listed-firm value declines with investor short-termism because firms shorten projects to attract short-horizon informed investors.
- Panel B plots the equilibrium project maturity of listed firms and the fraction of unlisted firms.
- As investor short-termism rises, project maturity falls.
- When short-termism becomes sufficiently high, some firms choose to remain unlisted.

Economic message: Investor short-termism can destroy the value of market monitoring. In the extreme, the benefit of listing disappears because competition for informed investors forces firms into excessively short projects.

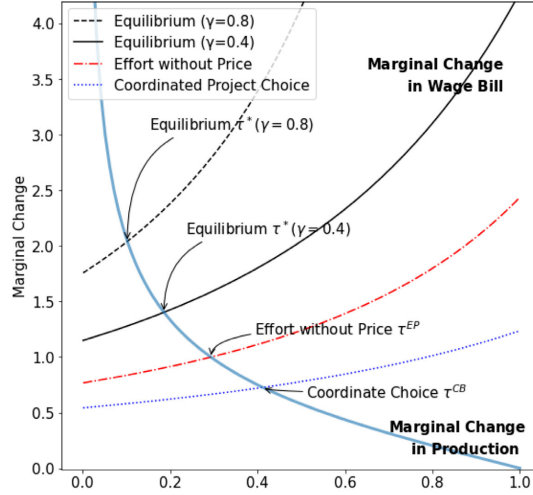


5.2 Figure 2: Equilibrium maturity versus benchmark maturities

Read:

- The upward-sloping black curves show marginal changes in the wage bill for different investor short-termism levels.
- The downward-sloping blue curve shows the marginal change in production from increasing maturity.
- Equilibrium maturity is where the wage-bill marginal change intersects the production marginal change.
- A higher γ shifts the equilibrium toward shorter maturity.
- The coordinated benchmark chooses longer maturity than the decentralized equilibrium.

Economic message: The figure visually shows why decentralized firms choose too-short maturities. Each firm reacts to the private benefit of attracting price informativeness, while a coordinated planner internalizes the externality across firms.



6 Short summary

- Informative stock prices can raise firm value by improving managerial incentives.
- Informed investors have limited horizons and prefer stocks whose information is resolved quickly.
- Firms choose short project maturities to attract informed investors and improve price informativeness.
- Informed trading capital is scarce, so attracting informed investors to one firm reduces informativeness for others.
- This creates strategic complementarity: one firm's short-termism induces others to shorten their projects.
- Equilibrium maturity is shorter than the coordinated benchmark.
- Shareholder value is lower in equilibrium than under coordinated project choices.
- When investor short-termism is high enough, some firms prefer not to list.
- Competition among firms and stronger agency problems can worsen the short-termism trap.
- Long-term investors can mitigate but do not necessarily eliminate the inefficiency.
- The central lesson is that stock-market monitoring can be valuable and yet still induce socially excessive short-termism.

7 Conclusion

7.1 Core conclusion

Dow, Han, and Sangiorgi show that stock market monitoring can generate a short-termism trap. Informative prices are valuable because they help incentivize managers, but firms compete for informed traders with limited horizons. Since informed trading capital is scarce, firms' attempts to attract informed investors are collectively wasteful and lead to inefficiently short project maturities.

7.2 Economic intuition

A firm can make its stock more attractive to short-horizon informed traders by choosing projects whose value is revealed sooner. This improves its own price informativeness and reduces its own managerial compensation

cost. But the informed investors it attracts come from other firms. When all firms shorten maturity, no firm gains aggregate informativeness, while all firms sacrifice the production benefits of longer-term projects.

7.3 Takeaway

The paper reconciles two views of stock markets: prices are useful for governance, but stock-market pressure can still be value destroying. The problem is not simply that managers are impatient or that investors are irrational. The problem is an informational externality in competition for informed attention.